PrO 15: Goodness-of-Fit Assessment of Observed Data Using Chi-Square Testing

A company monitors visits to four sections of its website: Dashboard, Reports, Settings, and Help. The company expects visitors to access all four sections equally. In a sample of 100 visits, the observed numbers are given below.

Website Section	Observed Visits
Dashboard	20
Reports	25
Settings	30
Help	25
Total	**100**

Using the **Chi-Square Goodness-of-Fit Test** at a 5% significance level, write a Python program to:

1. Calculate the expected number of visits for each section.
2. Calculate the chi-square test statistic and p-value.
3. Test whether the observed visits differ significantly from the expected equal distribution.

```
from scipy.stats import chisquare

observed = [20, 25, 30, 25]
expected = [25, 25, 25, 25]

chi, p = chisquare(observed, expected)

print("Chi-square value =", round(chi, 4))
print("P-value =", round(p, 4))

if p < 0.05:
    print("Reject H0: Observed visits differ significantly from the expected equal
distribution")
else:
    print("Accept H0: Observed visits do not differ significantly from the expected equal
distribution")
```

Output

Chi-square value = 2.0
P-value = 0.5724
Accept H0: Observed visits do not differ significantly from the expected equal distribution

A software company analyzes user clicks on four features of its mobile application: Home, Search, Profile, and Settings. The company expects users to access all four features equally. In a sample of 120 clicks, the observed frequencies are given below.

Mobile App Feature	Observed Clicks
Home	36
Search	24
Profile	30
Settings	30
Total	**120**

Using the **Chi-Square Goodness-of-Fit Test** at a 5% significance level, write a Python program to:

1. Calculate the expected frequency for each feature.
2. Calculate the chi-square test statistic and p-value.
3. Determine whether the observed clicks differ significantly from the expected equal distribution.